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SERVER-SIDE MODULE

CONTENTS

This module has the following files:

1. MODULE_SERVER_SIDE.doc
2. MODULE_SERVER_SIDE_MEDIA.zip

INTRODUCTION

A car sales company wants to sell cars to many types of people. not all people can buy a car with cash. So the company appoints you as a competent person to create an instalment cars platform.

The goal of this platform is for people to make car instalment transactions online.

DESCRIPTION OF PROJECTS AND TASKS

Registration involves the society platform (registering to the platform and applying for instalment), the company platform (giving instalment cars, viewing and accepting/rejecting registrations from the instalment), and the officer platform (updating company data, checking the validity of society data).

This module is divided into two phases. In the first phase, you will create a REST API. In the second phase, you will build a frontend application. You must use the provided frameworks:

- Laravel (with Sanctum)
- React (with react-router-dom and axios)
- Vue (with vue-router and axios)

Phase one – Society Car instalment REST API

Authentication

Before a society performs any activity on the society platform, they must be logged in using their **ID card number and password**. **Sanctum** should be used to create tokens. The generated token will be used to access all available endpoints.

Specifications :

Feature: A1 - Login and Logout as society: as a visitor, i want to login so that i can enter the system and apply for Instalment.

Scenario: Society Login

Request:

URL: [domain]/api/v1/auth/login

Method: POST

Header:

Body: { id_card_number: "SomeText", password: "SomeText" }

Response:

A1a - If login success

Header: Response code: 200

Body: {

```
"name": "Doni Rianto",
"born_date": "1974-10-22",
"gender": "male"|"female",
"address": "Ki. Raya Setiabudhi No. 790",
"token": "1|qgV2cTisoRVSwX6R2iPpxd..",
"regional": {
  "id": 1,
  "province": "DKI Jakarta",
  "district": "Central Jakarta"
}
```

}

A1b - If ID Card Number or Password incorrect

Header: Response code: 401

Body: { "message": "ID Card Number or Password incorrect" }

Scenario: Society Logout

Request:

URL: [domain]/api/v1/auth/logout

Params:

Method: POST

Header: { Authorization: Bearer xxxxxx }

Body:

Response:

A1c - If logout success

Header: Response code: 200

Body: { "message": "Logout success" }

A1d - If logout invalid token

Header: Response code: 401

Body: { "message": "Invalid token" }

Request Data Validation

To be able to apply for Instalment, the society must do validation data on the platform first. The society can view validity status and make validation requests. The society can only make one validation data request.

Specifications:

Feature: A2 - Request Data Validation: As a society, I want to request data validation so that I can apply for instalment when my data validation is accepted.

Scenario: Request data validations

Request:

URL: [domain]/api/v1/validation

Params:

Method: POST

Header:{ Authorization: Bearer xxxxxx }

Body: { job: "SomeText", job_description: "SomeText",
income: "SomeText" , reason_accepted: "SomeText" }

Response:

A2a - If request data validation success

Header: Response code: 200

```
Body: { "message": "Request data validation sent
successful" }
A2b - If invalid token
Header: Response code: 401
Body: { "message": "Unauthorized user" }
```

Scenario: Get society data validation

Request:

```
URL: [domain]/api/v1/validations
Params:
Method: GET
Header: { Authorization: Bearer xxxxxx }
Body:
```

Response:

```
A2c - If success
Header: Response code: 200
Body: {
  "validation": {
    "id": 1,
    "status": "pending"|"accepted"|"rejected",
    "job": "SomeText"|null,
    "job_description": "SomeText"|null,
    "income": "SomeText"|null,
    "reason_accepted": "SomeText"|null,
    "validator_notes": "SomeText"|null,
    "validator": { VALIDATOR }|null
  }
}
```

```
A2d - If invalid token
Header: Response code: 401
Body: {"message": "Unauthorized user"}
```

Instalment

When the validation data has been accepted by the validator and society will apply for instalment cars, the society can choose the cas based on the income that has been entered during data validation. List of available cars at each brand should also be displayed.

In the car's detail, it can calculate the price to be paid each month.

Specifications:

Feature: A3 - Instalment: As a society, I want to see a list of Instalment cars.

Scenario: Get all Instalment cars

Request:

URL: [domain]/api/v1/instalment_cars

Params:

Method: GET

Header: { Authorization: Bearer xxxxxx }

Body:

Response:

A3a - If success

Header: Response code: 200

Body: {

 "cars": [

 {

 "id": 1,

 "car": "Toyota FT 86",

 "brand": "Toyota",

 "price": "9000000000",

 "description": "Toyota FT 86 car is the best",

 "available_month": [

 {

 "month": 12,

 "description": "12 Months",

 },

 ...,

]

```

        },
        ...,
    ]
}
A3b - If invalid token
Header: Response code: 401
Body: { "message": "Unauthorized user" }

```

Scenario: Get Instalment detail by instalment ID

Request:

```

URL: /api/v1/instalment_cars/<id>
Params: { id }
Method: GET
Header: { Authorization: Bearer xxxxxx }
Body:

```

Response:

```

A3c - If success
Header: Response code: 200
Body: {
    "instalment": {
        "id": 1,
        "car": "Toyota FT 86",
        "brand": "Toyota",
        "price": "9000000000",
        "description": "Toyota FT 86 car is the
                        best",
        "available_month": [
            {
                "month": 12,
                "description": "12 Months",
            },
            ...,
        ]
    },
}

```

A3d - If invalid token

Header: Response code: 401

Body: { "message": "Unauthorized user" }

Applying for instalment

The society can apply for an Instalment if the validation data has been accepted by the validator.

Application for an Instalment can only be once.

Instalment ID and position must be filled.

Specifications:

Feature: A4 - Applying for Instalment: as a society, i want to applying for Instalment.

Scenario: Applying for instalment

Request:

URL: /api/v1/applications

Params:

Method: POST

Header: { Authorization: Bearer xxxxxx }

Body: { instalment_id: 1, months: 12, notes : "SomeText" }

Response:

A4a - If success

Header: Response code: 200

Body: { "message": "Applying for Instalment successful" }

A4b - If invalid token

Header: Response code: 401

Body: { "message": "Unauthorized user" }

A4c - If the society validation data hasn't accepted by validator

Header: Response code: 401

Body: { "message": "Your data validator must be accepted by validator before" }

A4d - If invalid fields

Header: Response code: 401

```
Body: {
  "message": "Invalid field",
  "errors": {
    "instalment_id": [
      "The instalment id field is required."
    ],
    "months": [
      "The months field is required."
    ]
  }
}
```

A4e - If have been 2x applying

Header: Response code: 401

Body: { "message": "Application for a instalment can only
be once" }

Scenario: Get all of society instalment applications

Request:

URL: [domain]/api/v1/applications

Params:

Method: GET

Header: { Authorization: Bearer xxxxxx }

Body: Response:

A4f - If success

Header: Response code: 200

```
Body: {
  "instalments": [
    {
      "id": 1,
      "car": "Toyota FT 86",
      "brand": "Toyota",
      "price": "9000000000",
      "description": "Toyota FT 86 car is the  
best",
      "applications": [
```

```

    {
      "month": "12",
      "nominal": "75000000",
      "apply_status":
        "pending"|"accepted"|"rejected",
      "notes": "I want this cars",
    },
    ...,
  ]
},
...,
]
}

```

A4g - If invalid token

Header: Response code: 401

Body: { "message": "Unauthorized user" }

POSTMAN Collection

You have to create a postman collection to check the API that has been created. An example is as follows:

- Installment Cars - Authentication - POST A1a - Login success - POST A1b - Login failed - POST A1c - Logout success - POST A1d - Logout failed - Request Data Validation - POST A2a - Request validation success - POST A2b - Request validation invalid token - Get the Validation - GET A2c - Get validation success - GET A2d - Get validation invalid token - Installment - GET A3a - Get all Installment cars - GET A3b - Get all Installment cars invalid token	- Installment Detail - GET A3c - Get detail installment cars - GET A3d - Get detail Installment cars invalid token - Applying for installment - POST A4a - Applying for installment success - POST A4b - Applying for installment invalid token - POST A4c - If the society validation hasn't accepted by validator - POST A4d - Invalid field - POST A4e - If 2nd apply - Get all installment applications - GET A4f - Get all installment applications success - GET A4g - Get all installment applications invalid token
--	--

Phase two - Society instalment Cars Platform (Frontend)

In this phase, you can use the gui template provided in the media files to build a society platform. Use the REST API following the REST API you created before.

Implement REST API into the frontend. All response formats must be in JSON form.

Society login and logout

To be able to enter the system, the society must first login using the ID card number and password. When the society has entered the system, the logout button will appear, and the society can exit the system by clicking the logout button

Specifications:

Feature: B1 - Society login and logout: as a visitor, i want to login into society instalment cars platform and logout to exit the system.

Scenario: B1a - Society login success

Given I am on the login page

And I enter correct id_card_number and password

Then I redirected to society dashboard page

Scenario: B1b - Society login fails

Given I am on the login page

And I enter wrong id_card_number or password

Then I stay in the login page and I see error message "ID Card Number or Password incorrect"

Scenario: B1c - Society logout success

Given I am on the any page that has been logged in

And I press logout button

Then I get message "logout success" and redirected to login page

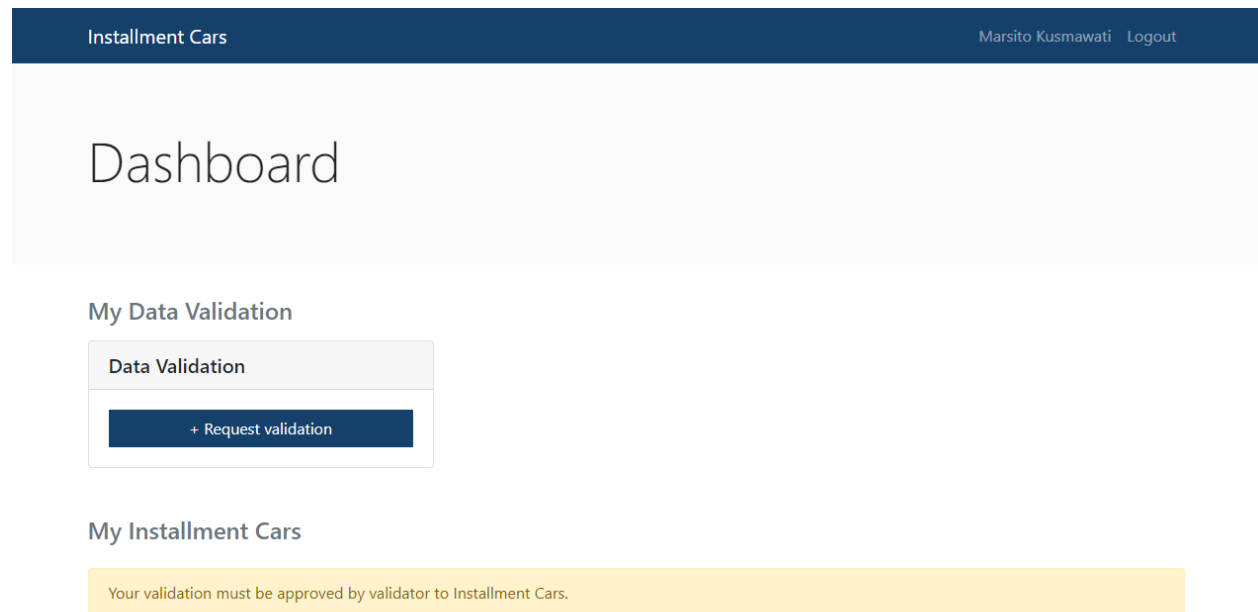
And I cannot reach the previous page by going back

Society Dashboard

In the dashboard page, there are two sections, data validation and instalment cars section. where the society must request a data validation and the data validation is approved by the validator to be able to register for instalment cars application.

Specifications:

Feature: B2 - Society Dashboard: as a society, i want to see data validation info and my instalment applications list in the dashboard page.



Scenario: B2a - Data validation section is displayed

Given I am on dashboard page

Then I see the data validation section with "My Data validation" title

Scenario: B2b - Link to request data validation appears when society hasn't data validation before

Given I am on dashboard page

And I haven't request data validation before

Then I see request data validation link in the data validation section

Scenario: B2c - Show data validation info

Data Validation	
Status	Accepted
Job	Programmer
Income/Month	Rp. 3.000.000
Validator	Usman M.Ti
Validator Notes	siap kerja

Given I am on dashboard page

And my request data validation is sent or accepted/rejectedd by validator

Then I see data validation info in the data validation section

Scenario: B2d - instalment cars application list section is displayed

Given I am on dashboard page

Then I see the instalment cars application list section with "instalment applications" title

Scenario: B2e - Show instalment cars application info message

Given I am on dashboard page

And I haven't request data validation before

Then I see "Your data validation must be approved by validator to get the applying instalment" in the instalment cars application list section

Scenario: B2f - The car instalment application is displayed when society data validation have been accepted by validator

My Installment Cars

+ Add Installment Cars

Given I am on dashboard page

And I my request data validation has been accepted by validator

Then I see the instalment cars application with link is displayed in the instalment cars application list section

Toyota FT 86	
Description	Toyota FT 86 car is the best
Price	Rp. 900.000.000
Installment	12 Months Pending
Apply Date	Desember 12, 2022
Notes	I want this one, because i am rich.

Given I am on dashboard page

And my instalment cars application is finished

Scenario: B2h - The instalment applications show instalment cars application info correctly when instalment cars application is finished

Given I am on dashboard page

Then I see instalment cars application info (cars, total months, apply date, notes) in the instalment cars application box

Request Data validation

In the data validation request page, the society can fill in the job, job description and income if they have by selecting "Yes, I have" in the dropdown. Are you working? question and they can send a request if the reason accepted has been filled.

Specifications :

Feature: B3 - Society Data validation: as a society, I want to request data validation.

Request Data Validation

Are you working? Yes, I have ▾

Your Job

describe what you do in your job

Income (Rp)

Reason Accepted

Explain why you should be accepted

Send Request

Scenario: B3a - Data validation form is displayed correctly

When I press "Send Request" button in the dashboard page
Then I see the data validation form (job, job description, income, reason accepted) is displayed

Scenario: B3b - Request data validation successful

Given I am on request data validation page
And I press request button
Then the request data validation is sent, the "Request data validation successful" message is appear, and I redirected to dashboard page

instalment Cars List

When the society registers for instalment cars application by pressing the "Add instalment cars application" button on the dashboard page, the society is directed to the list instalment Cars page to select instalment Cars. In the item instalment Cars, display the applying instalment available in those cars correctly.

Specifications :

Feature: B4 - instalment Cars List: as a society, I want to see a list of all instalment Cars.

Installment Cars			Marsito Kusmawati	Login
Cars				
List of Cars				
Toyota FT 86 Toyota FT 86 car is the best		Available Month 12 Months, 24 Months, 48 Months	Detail	
Nissan Livina Family Cars for everyone		Available Month 12 Months, 24 Months	Vacancies have been submitted	
Toyota FT 86 Toyota FT 86 car is the best		Available Month 12 Months, 24 Months, 48 Months	Detail	

Scenario: B4a - Show list instalment Cars

Given I am on instalment Cars list page

Then I see a list of instalment Cars

Scenario: B4b - Show information for each instalment Cars

Given I am on instalment Cars page

Then I see instalment information (car name, description, available month) in each instalment item.

Scenario: B4c - instalment item unavailable when the instalment has been apply before.

Detail instalment

In the instalment detail page, there are details of instalment shown like car name, address, car description, available months. Society can apply instalment by filling months and notes for application.

Toyota FT 86

Brand : Toyota

Description

Toyota FT 86 car is the best

Price : **Rp. 900.000.000**

Select Months

12 Months

Nominal/Month : **Rp. 10.000.000**

Notes

Explain why your installment should be approved

Apply

Specifications :

Feature: B5 - Detail Instalment: as a society, I want to see details of Instalment and see available months.

Scenario: B5a - Information displays correctly

Given I am on instalment detail page

Then I see the section with available months and calculate the price to be paid each month.

Scenario: B5b - Can apply for a instalment if the income equal or exceed the calculation per month

Given I am on instalment detail page

Then I see available months and calculate the price to be paid each month

Scenario: B5c - Register instalment cars application

Given I am on instalment detail page

I choose total months

I fill the Notes for instalment field

And I press "Apply" button

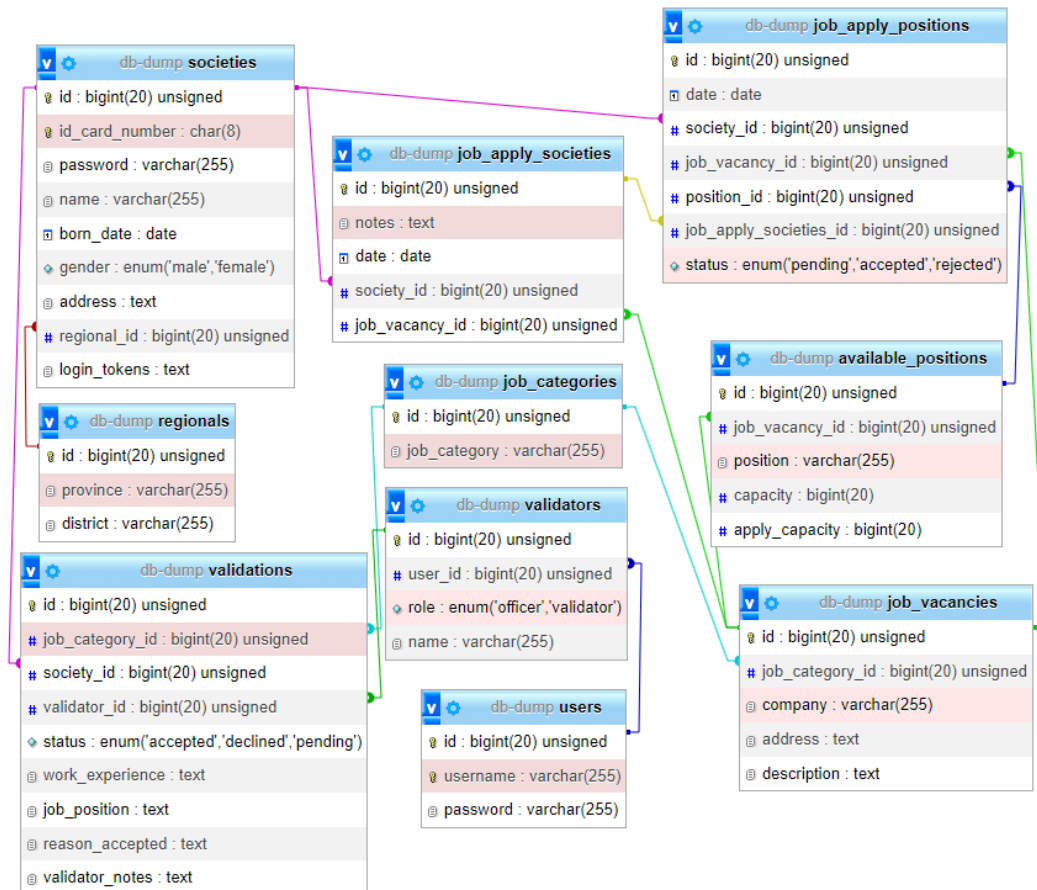
Then the success or error message is appear and I redirected to dashboard page

Notes:

- For each response from the REST API must be displayed.
- Build the frontend into production mode first before your frontend project is pushed and the REST API must be consumed from your API before.

ERD

You can make your own ERD according to your needs. bellow is an example of ERD:



INSTRUCTION FOR COMPETITORS

- You may create additional endpoints if necessary, but you must fulfil all of the requirements listed above.
- You can add additional fields to the response body if needed, but ensure that all required fields are included and the structure is displayed correctly.
- Create a root folder called **XX_SERVER_MODULE** in your local computer, where **XX** is your computer number.
- **Created Postman Collection, Database (db-dump.sql) and ER diagram (db-diagram.pdf)** should be exported and saved under the root folder **<host>/XX_SERVER_MODULE**
- REST API should be reached on **<host>/XX_SERVER_MODULE/BACKEND** where **xx** is PC number. No port and no /public suffix path.
- Frontend app should be reached on **<host>/XX_SERVER_MODULE/FRONTEND**.
- Zip root folder **XX_SERVER_MODULE** and submit to the submission page. Make sure to exclude the **vendor** and **node_modules** folder when zipping files.

CLIENT SIDE MODULE

CONTENTS

This module has the following files:

1. CLIENT SIDE MODULE.docx
2. CLIENT SIDE MODULE.pdf
3. CLIENT SIDE MODULE MEDIA.zip

PROJECT OVERVIEW

A client needs a visual map that can display various locations and routes to each location. This map should be interactive, allowing users to zoom, pan, and click on markers to view detailed information. The website needs to be compatible with modern web browsers and mobile devices.

This module is a 3-hour competition task. You are required to develop a web-based application using only HTML, CSS, and JavaScript. The core functionality revolves around an interactive map of Indonesia, allowing users to add, connect, and manage pinpoint locations, as well as find routes between them. You are allowed to improve the usability and user interface for a better user experience.

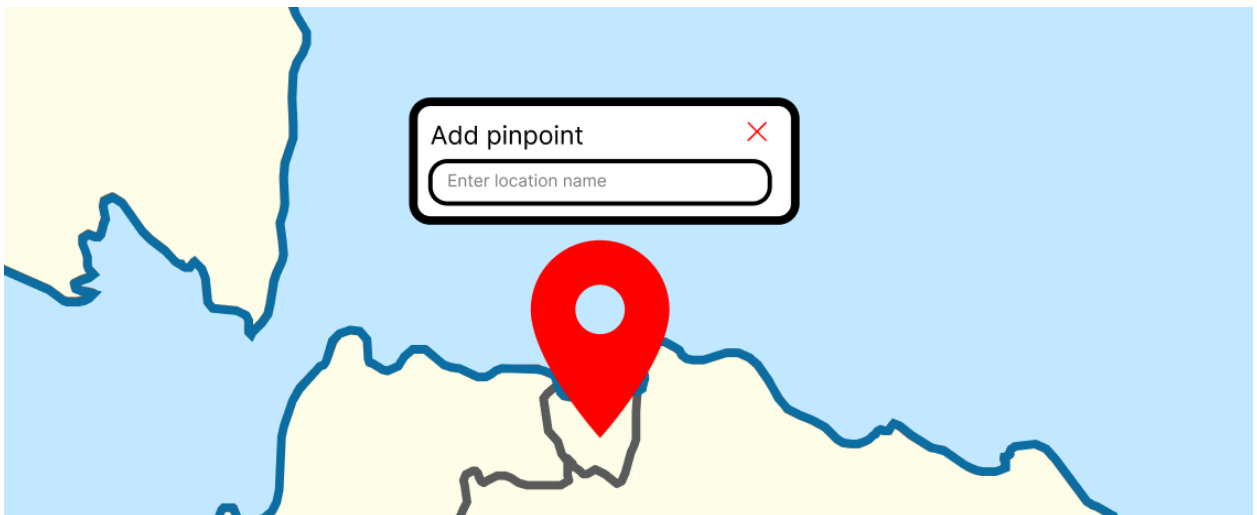
Loading the Map

You are provided with an SVG image of the Indonesia map. The map should be loaded into the page full size zoomed-in without leaving any blank space.



Adding Pinpoint Location

Double-click in the map will show a popup containing an input for location name to add a pinpoint in the map. Clicking the enter button will submit the form.



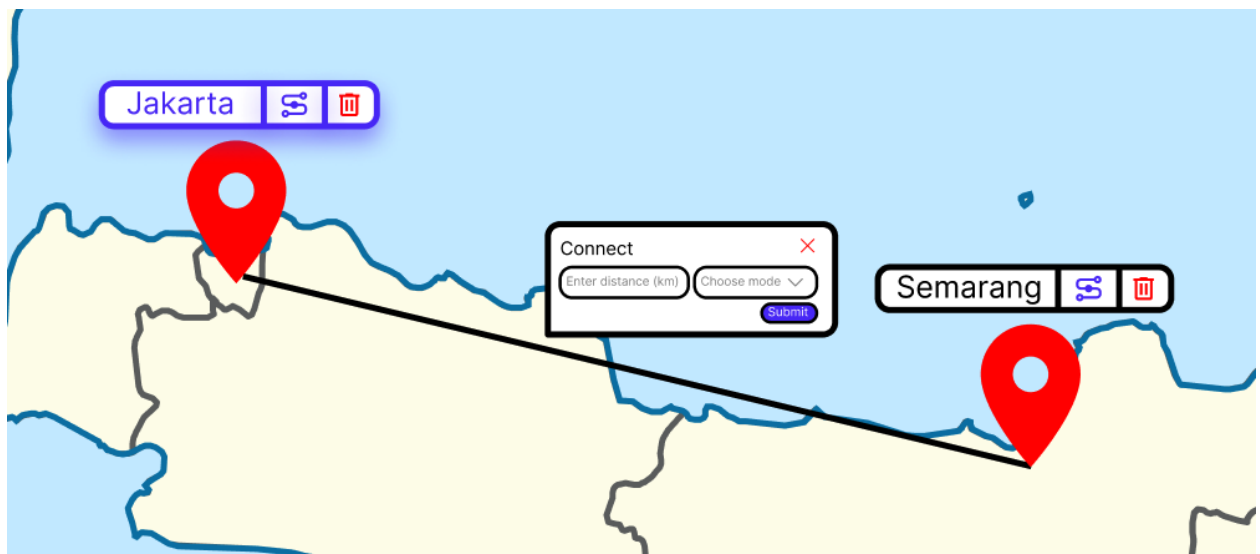
If the pinpoint is added, it should show the pinpoint name, a button to connect the pinpoint, and a button to delete it. A pinpoint should be marked with a red map-pin icon.



The created pinpoint should be persistent. Reloading the page will not remove the pinpoints.

Connect Locations

If the user clicked the connect icon, the pinpoint label shows signs that the pinpoint is in the state of connecting (e.g. glow, changed colors). Then, click the target pinpoint and the *connect popup* will show.



In this form, users need to enter the distance in kilometers and the transportation mode. The transportation modes consist of:

1. Train

- Line color: #33E339
- Speed: 120 km/h
- Cost: Rp500/km

2. Bus

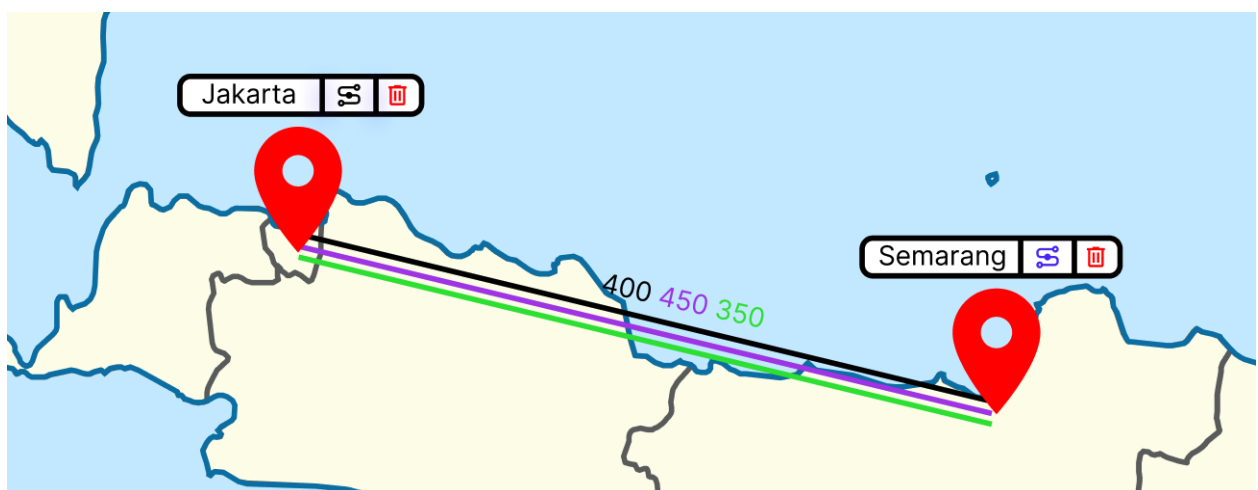
- Line color: #A83BE8
- Speed: 80 km/h
- Cost: Rp100/km

3. Airplane

- Line color: #000000
- Speed: 800 km/h
- Cost: Rp1.000/km



After submit, **a line will appear connecting both pinpoint locations** with the distance present in the middle of the line.



Users can add up to all transportation modes for each connection. Each connection can have a different distance and the distances (in km) will show in the center of the line.



Pan and zoom should be working correctly. Users should be able to zoom-in and zoom-out the map by clicking **CTRL+Scroll** or **CTRL+(+) and CTRL+(-)**. Zoom-in and zoom-out functionality should follow the cursor position.

In a zoomed map, users should be able to **move the map left and right by holding click and moving the cursor**.

Find Route

The Find Route modal should always exist on the left side of the browser viewport. Users should be able to input “*from pinpoint*” and “*to pinpoint*” with the exact keyword name. If one of the inputs contains invalid pinpoint name, **the search button should be disabled**.

Clicking the search button will show **up to 10 possible routes** maximum, sorted by **fastest duration** or the **cheapest cost**. The order should be **the fastest** by default.

In each route, **show the route name, the transportation steps, the duration, and the total cost**.

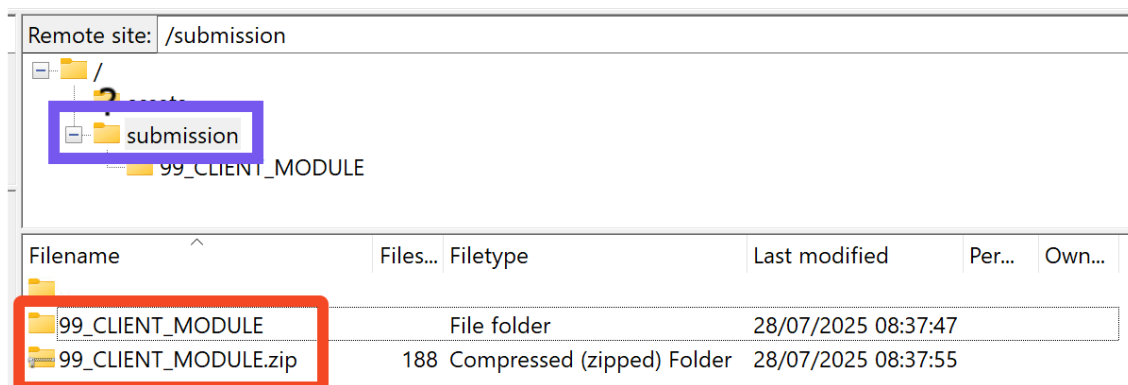


Removing Pinpoint and Connection

To remove a connection between pinpoints, the **user should be able to click the line, and press the del/backspace button**. A pinpoint can be removed by clicking the trash can button in the label, removing all of its connection.

INSTRUCTIONS TO THE COMPETITORS

1. This project will be assessed with either Firefox Developer Edition or Google Chrome web browser.
2. You may provide a README file for executing guide if necessary.
3. Create a root folder called **XX_CLIENT_SIDE_MODULE** in your local computer, where **XX** is your **computer number**.
4. Place your works inside the root folder. Make sure your works are working well when directly opening the **index.html** file.
5. Upload your **XX_CLIENT_SIDE_MODULE** to the FTP server inside “**submission**” folder (Please look at the example below).
6. Zip your root folder **XX_CLIENT_SIDE_MODULE** and upload to FTP server (Please look at the example below).



OTHERS

#	Sub-Criteria	Marks
1	Map functionalities	8.5
2	Pinpoint control	4.5
3	Pinpoint connections	7
4	Find Routes	10
Total		30